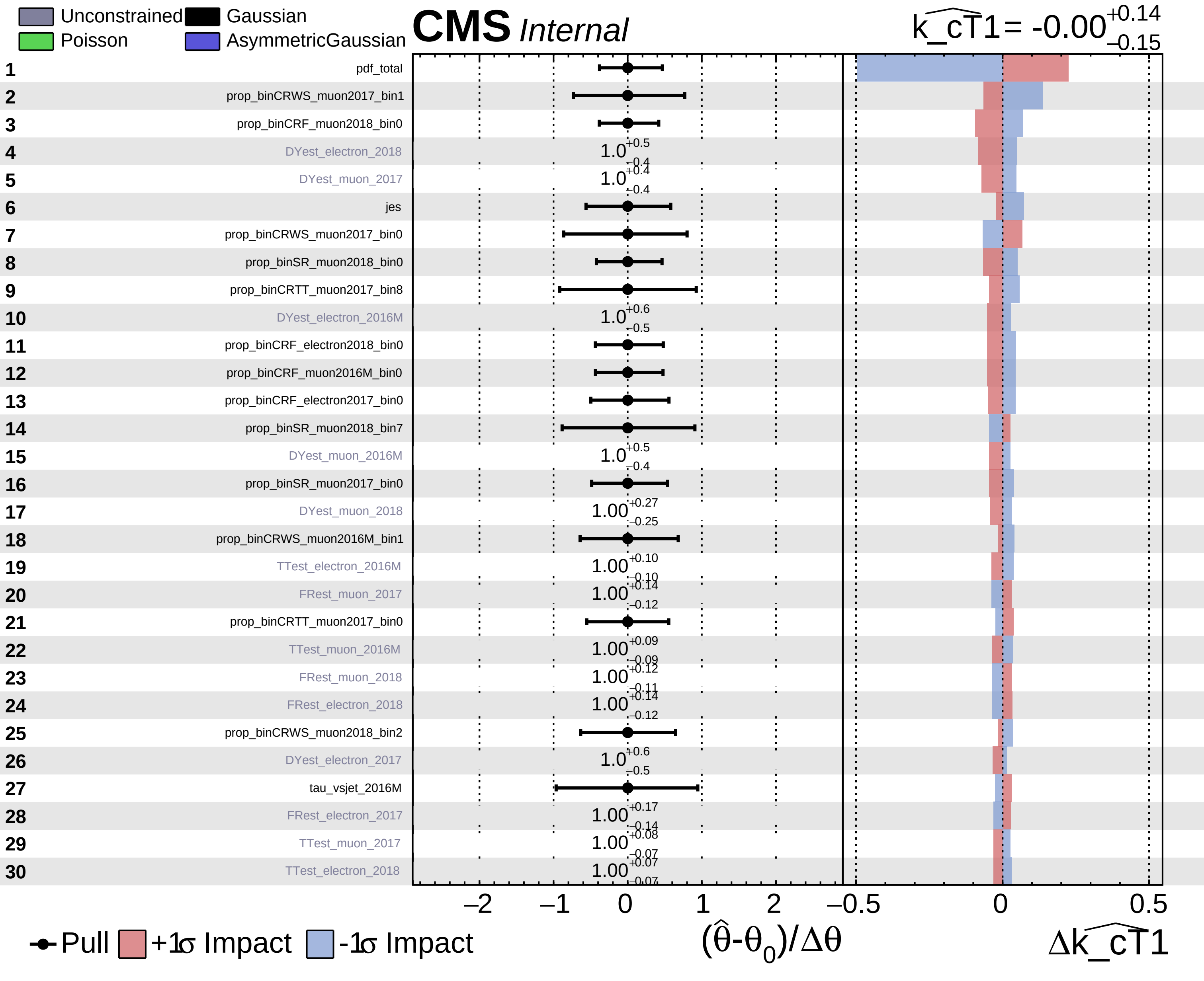
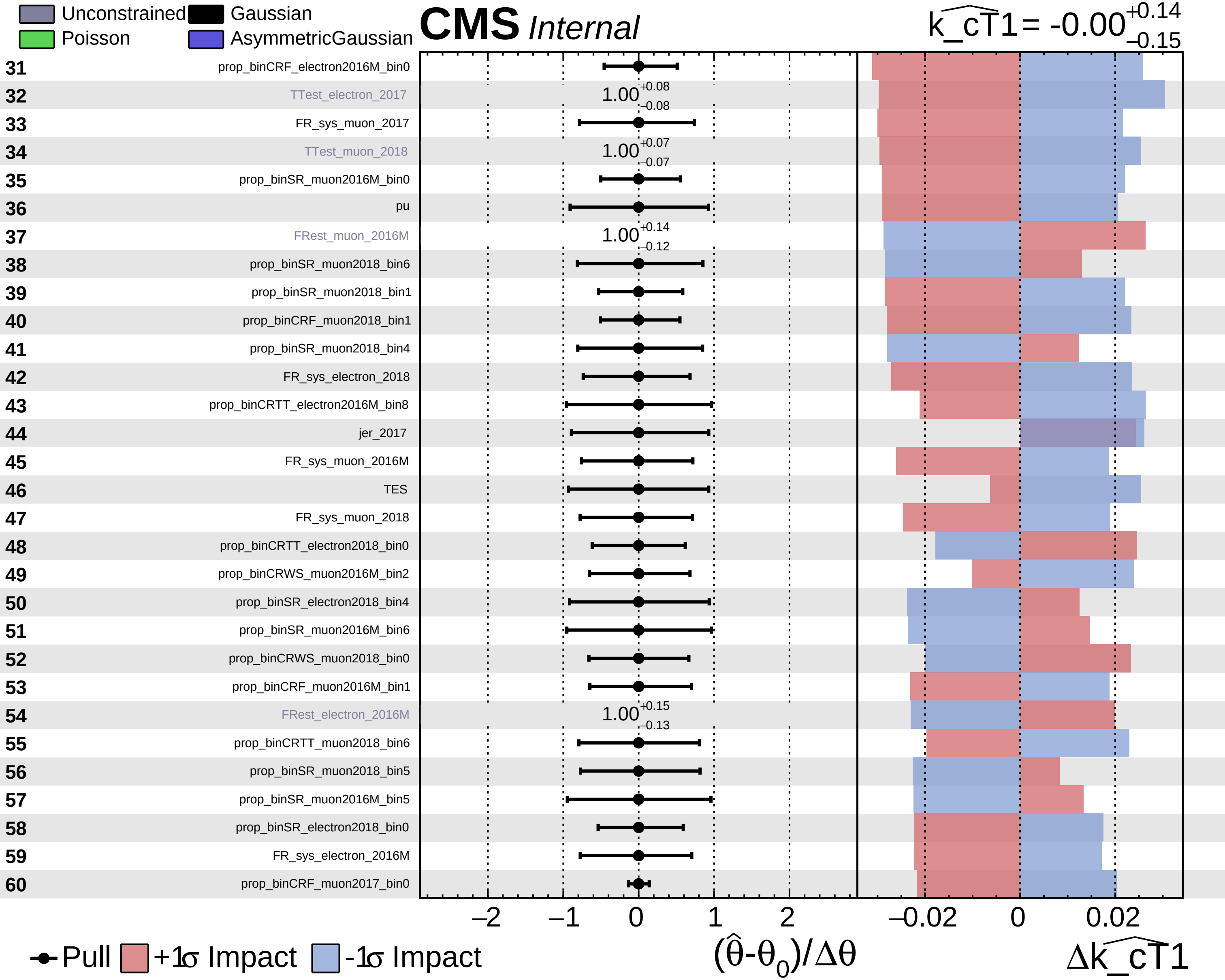
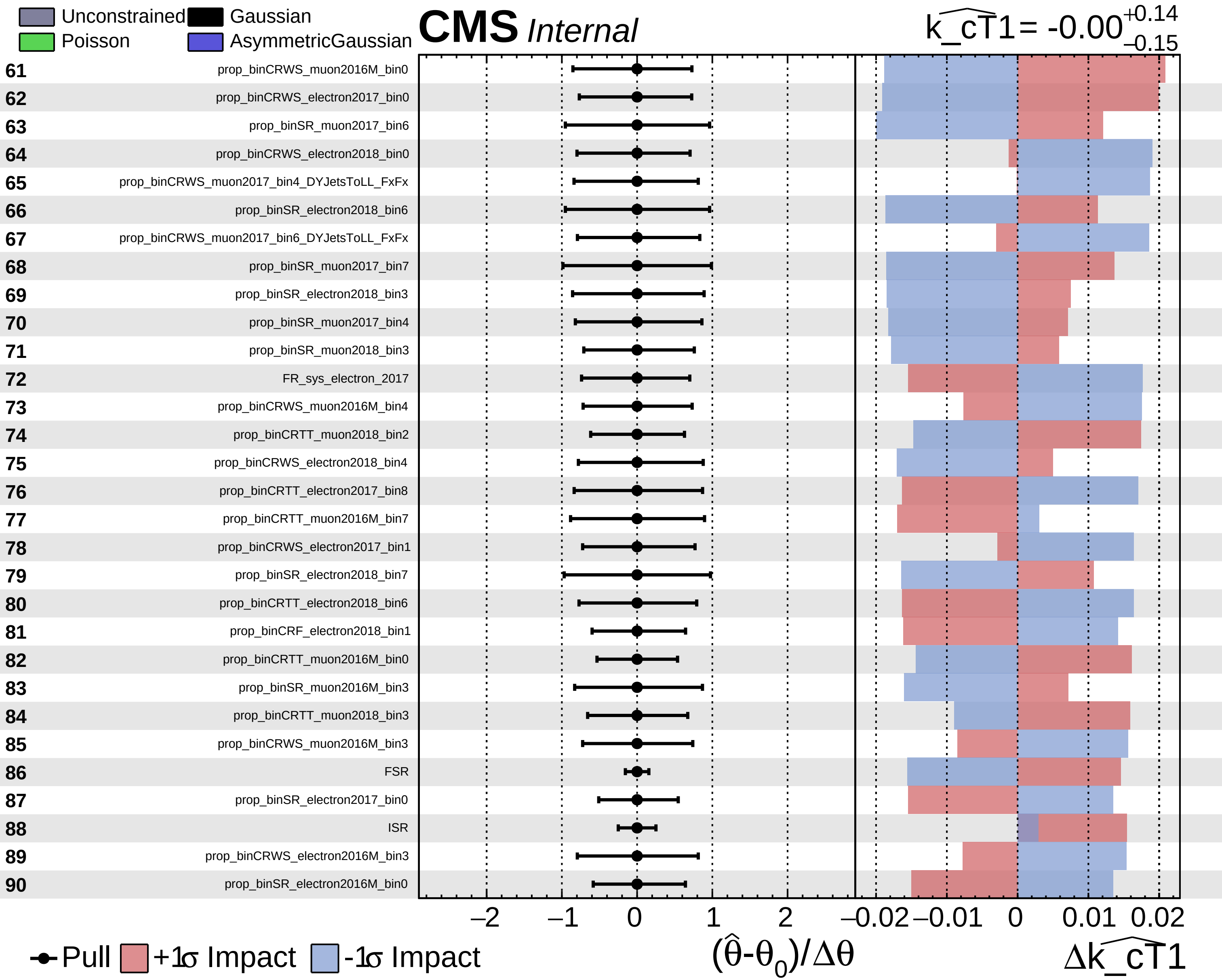


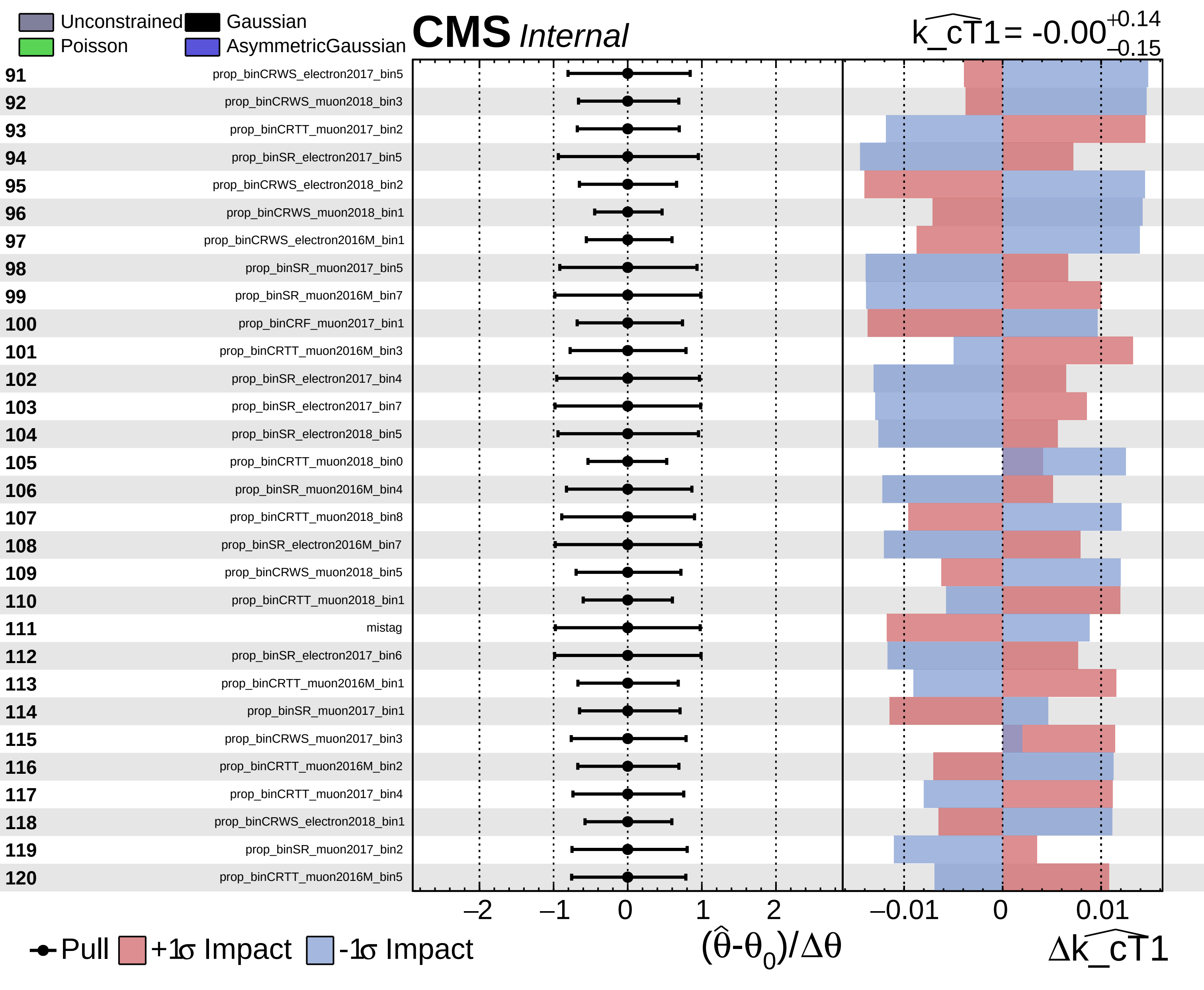
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = -0.00 +0.14 −0.15
1 pdf_total
2 prop_binCRWS_muon2017_bin1
3 prop_binCRF_muon2018_bin0
4 DYest_electron_2018 1.0 +0.5 −0.4
5 DYest_muon_2017 1.0 +0.4 −0.4
6 jes
7 prop_binCRWS_muon2017_bin0
8 prop_binSR_muon2018_bin0
9 prop_binCRTT_muon2017_bin8
10 DYest_electron_2016M 1.0 +0.6 −0.5
11 prop_binCRF_electron2018_bin0
12 prop_binCRF_muon2016M_bin0
13 prop_binCRF_electron2017_bin0
14 prop_binSR_muon2018_bin7
15 DYest_muon_2016M 1.0 +0.5 −0.4
16 prop_binSR_muon2017_bin0
17 DYest_muon_2018 1.00 +0.27 −0.25
18 prop_binCRWS_muon2016M_bin1
19 TTest_electron_2016M 1.00 +0.10 −0.10
20 FRest_muon_2017 1.00 +0.14 −0.12
21 prop_binCRTT_muon2017_bin0
22 TTest_muon_2016M 1.00 +0.09 −0.09
23 FRest_muon_2018 1.00 +0.12 −0.11
24 FRest_electron_2018 1.00 +0.14 −0.12
25 prop_binCRWS_muon2018_bin2
26 DYest_electron_2017 1.0 +0.6 −0.5
27 tau_vsjet_2016M
28 FRest_electron_2017 1.00 +0.17 −0.14
29 TTest_muon_2017 1.00 +0.08 −0.07
30 TTest_electron_2018 1.00 +0.07 −0.07
−2 −1 0 1 2
−0.5 0 0.5
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1



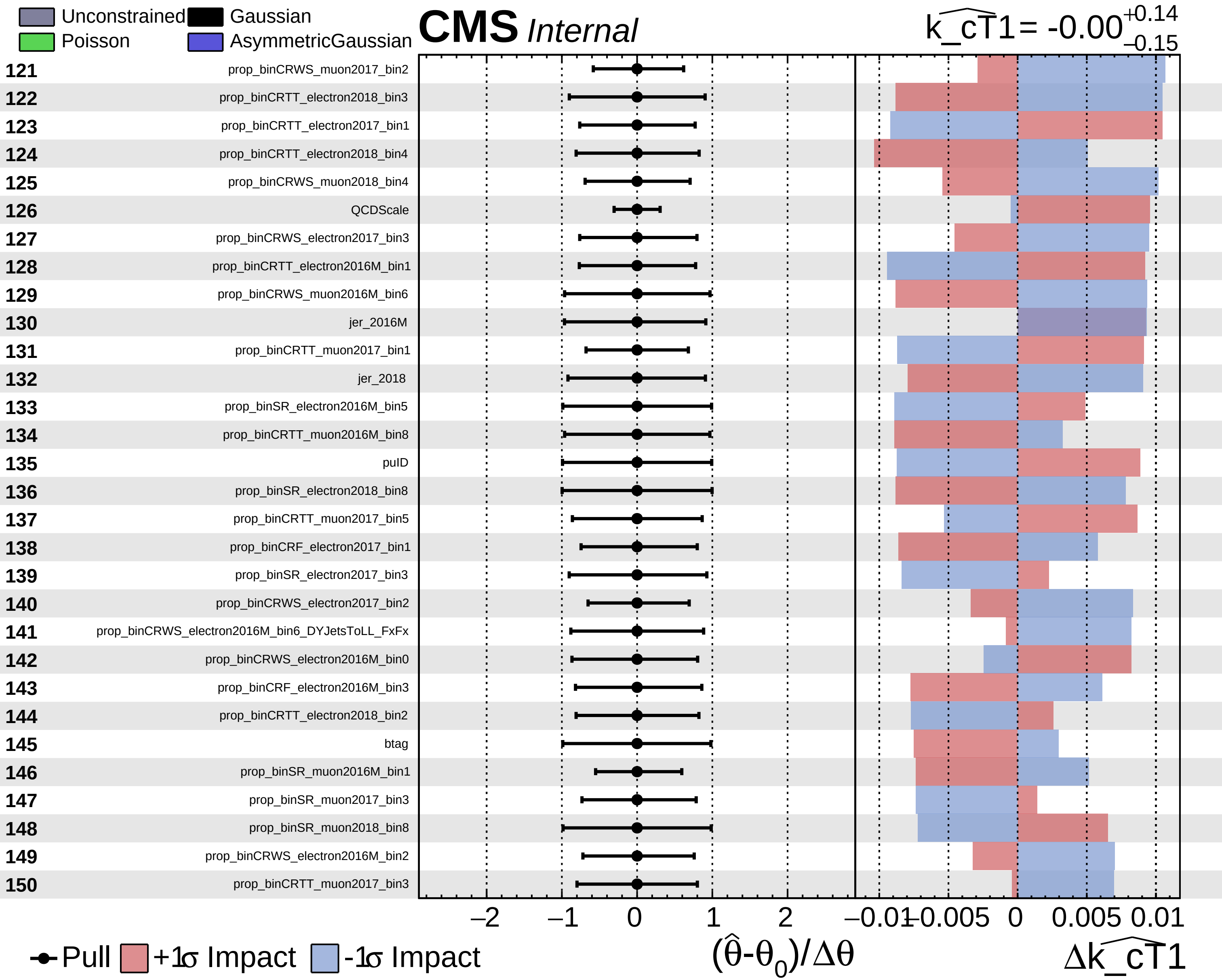
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT1} = -0.00^{+0.14}_{-0.15}$
31 prop_binCRF_electron2016M_bin0
32 TTest_electron_2017
$1.00^{+0.08}_{-0.08}$
33 FR_sys_muon_2017
34 TTest_muon_2018
$1.00^{+0.07}_{-0.07}$
35 prop_binSR_muon2016M_bin0
36 pu
37 FRest_muon_2016M
$1.00^{+0.14}_{-0.12}$
38 prop_binSR_muon2018_bin6
39 prop_binSR_muon2018_bin1
40 prop_binCRF_muon2018_bin1
41 prop_binSR_muon2018_bin4
42 FR_sys_electron_2018
43 prop_binCRTT_electron2016M_bin8
44 jer_2017
45 FR_sys_muon_2016M
46 TES
47 FR_sys_muon_2018
48 prop_binCRTT_electron2018_bin0
49 prop_binCRWS_muon2016M_bin2
50 prop_binSR_electron2018_bin4
51 prop_binSR_muon2016M_bin6
52 prop_binCRWS_muon2018_bin0
53 prop_binCRF_muon2016M_bin1
54 FRest_electron_2016M
$1.00^{+0.15}_{-0.13}$
55 prop_binCRTT_muon2018_bin6
56 prop_binSR_muon2018_bin5
57 prop_binSR_muon2016M_bin5
58 prop_binSR_electron2018_bin0
59 FR_sys_electron_2016M
60 prop_binCRF_muon2017_bin0
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$



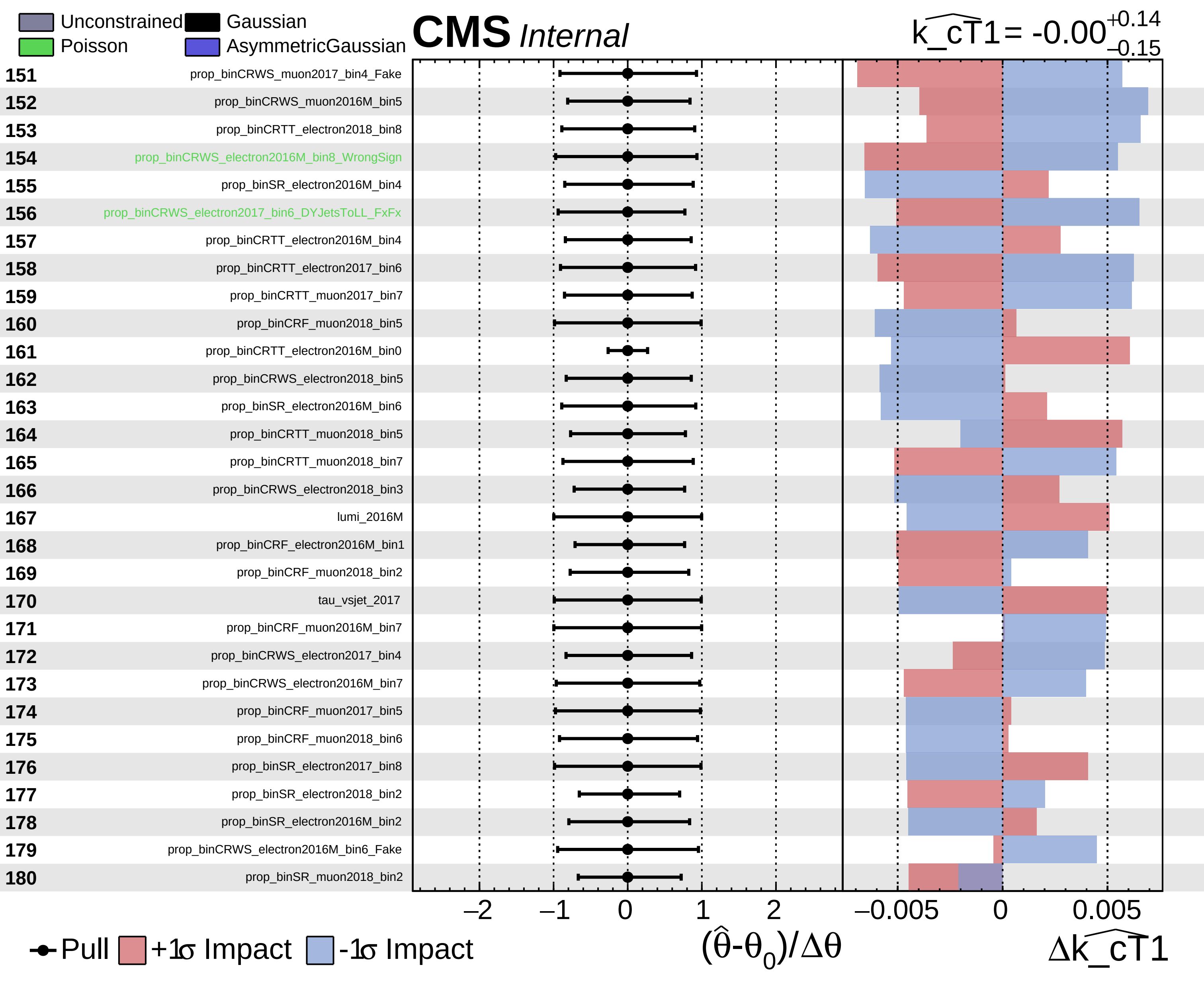
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = -0.00 +0.14 −0.15
61 prop_binCRWS_muon2016M_bin0
62 prop_binCRWS_electron2017_bin0
63 prop_binSR_muon2017_bin6
64 prop_binCRWS_electron2018_bin0
65 prop_binCRWS_muon2017_bin4_DYJetsToLL_FxFx
66 prop_binSR_electron2018_bin6
67 prop_binCRWS_muon2017_bin6_DYJetsToLL_FxFx
68 prop_binSR_muon2017_bin7
69 prop_binSR_electron2018_bin3
70 prop_binSR_muon2017_bin4
71 prop_binSR_muon2018_bin3
72 FR_sys_electron_2017
73 prop_binCRWS_muon2016M_bin4
74 prop_binCRTT_muon2018_bin2
75 prop_binCRWS_electron2018_bin4
76 prop_binCRTT_electron2017_bin8
77 prop_binCRTT_muon2016M_bin7
78 prop_binCRWS_electron2017_bin1
79 prop_binSR_electron2018_bin7
80 prop_binCRTT_electron2018_bin6
81 prop_binCRF_electron2018_bin1
82 prop_binCRTT_muon2016M_bin0
83 prop_binSR_muon2016M_bin3
84 prop_binCRTT_muon2018_bin3
85 prop_binCRWS_muon2016M_bin3
86 FSR
87 prop_binSR_electron2017_bin0
88 ISR
89 prop_binCRWS_electron2016M_bin3
90 prop_binSR_electron2016M_bin0
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT1



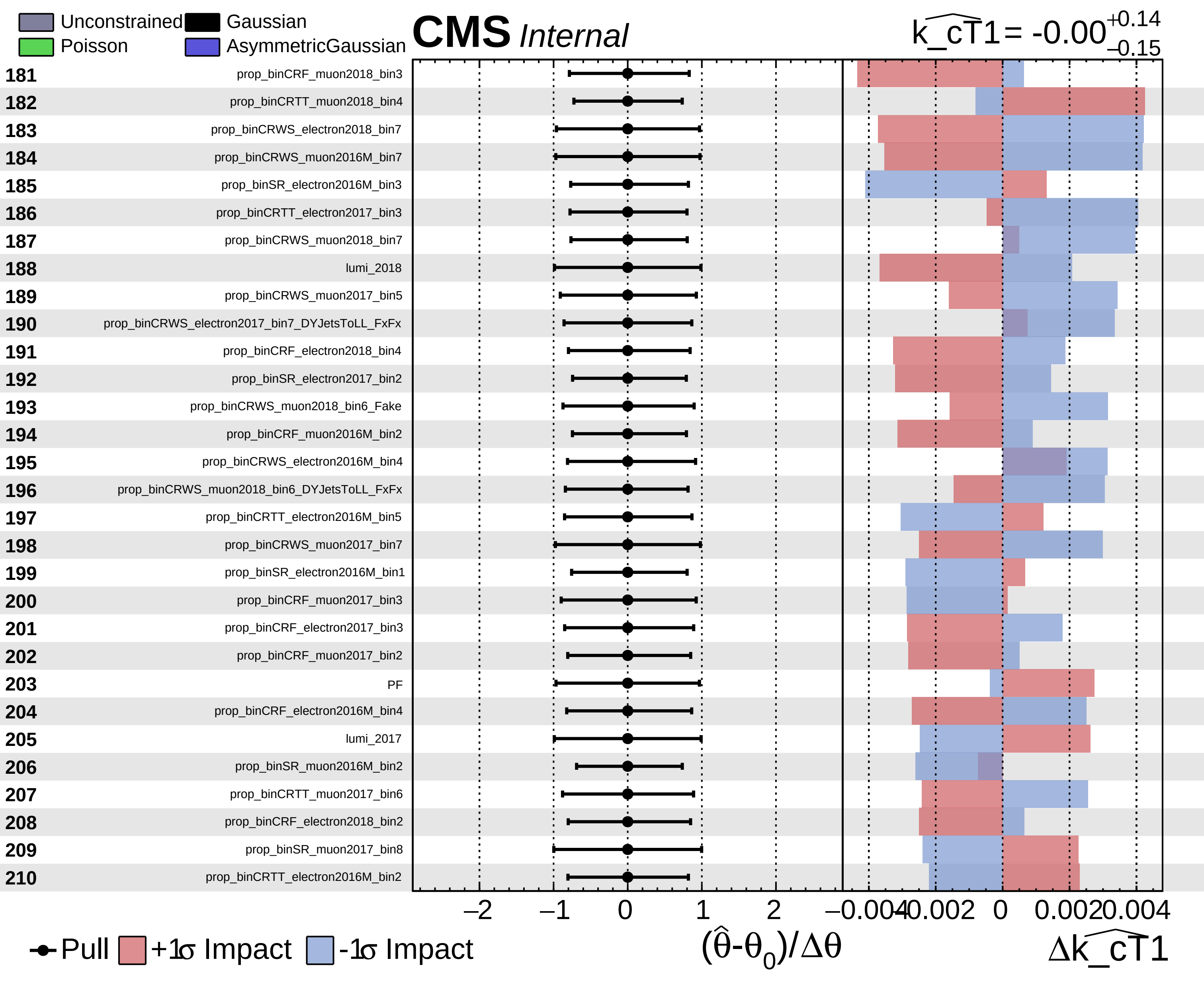
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT1} = -0.00^{+0.14}_{-0.15}$
91 prop_binCRWS_electron2017_bin5
92 prop_binCRWS_muon2018_bin3
93 prop_binCRTT_muon2017_bin2
94 prop_binSR_electron2017_bin5
95 prop_binCRWS_electron2018_bin2
96 prop_binCRWS_muon2018_bin1
97 prop_binCRWS_electron2016M_bin1
98 prop_binSR_muon2017_bin5
99 prop_binSR_muon2016M_bin7
100 prop_binCRF_muon2017_bin1
101 prop_binCRTT_muon2016M_bin3
102 prop_binSR_electron2017_bin4
103 prop_binSR_electron2017_bin7
104 prop_binSR_electron2018_bin5
105 prop_binCRTT_muon2018_bin0
106 prop_binSR_muon2016M_bin4
107 prop_binCRTT_muon2018_bin8
108 prop_binSR_electron2016M_bin7
109 prop_binCRWS_muon2018_bin5
110 prop_binCRTT_muon2018_bin1
111 mistag
112 prop_binSR_electron2017_bin6
113 prop_binCRTT_muon2016M_bin1
114 prop_binSR_muon2017_bin1
115 prop_binCRWS_muon2017_bin3
116 prop_binCRTT_muon2016M_bin2
117 prop_binCRTT_muon2017_bin4
118 prop_binCRWS_electron2018_bin1
119 prop_binSR_muon2017_bin2
120 prop_binCRTT_muon2016M_bin5
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$

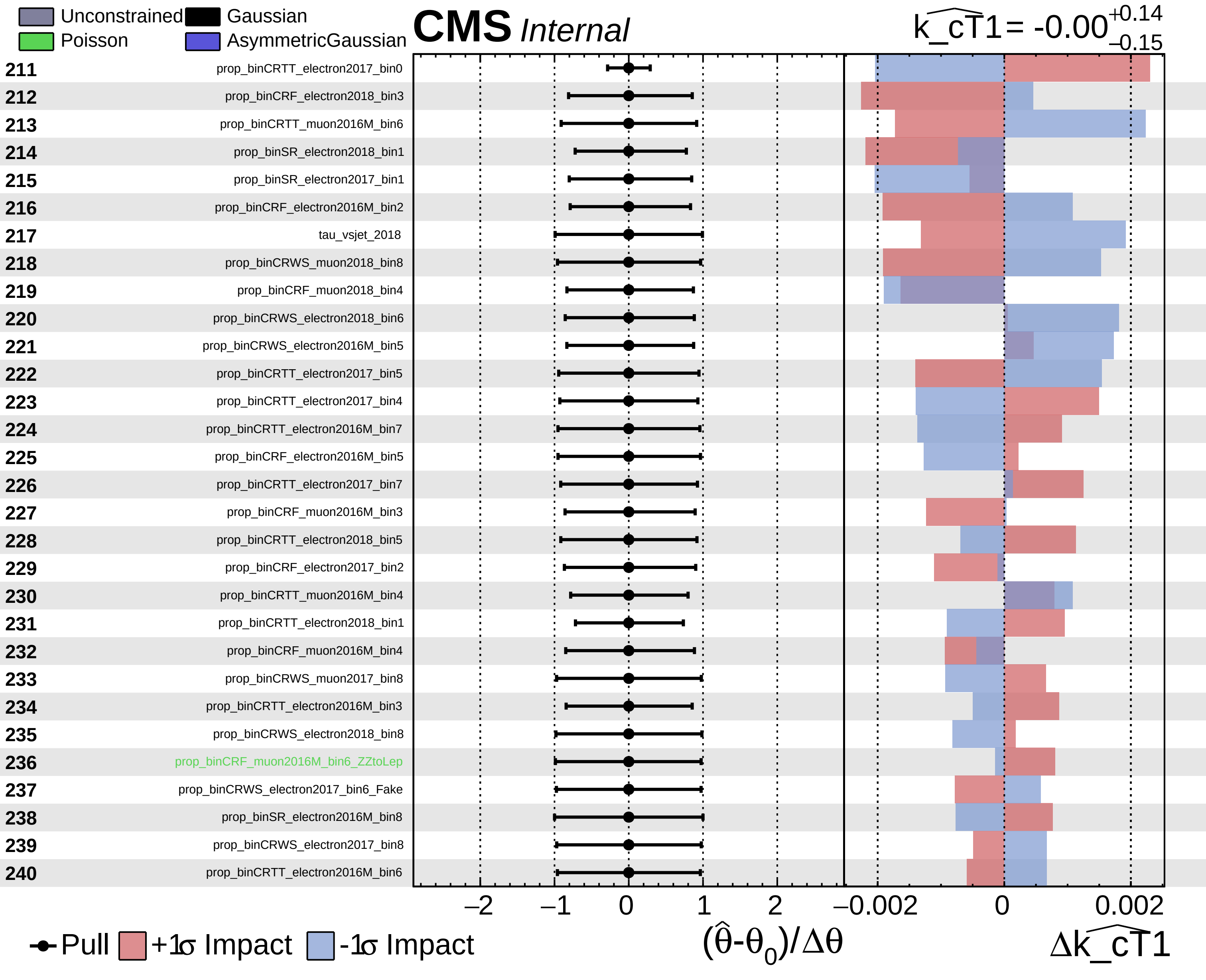


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT1} = -0.00^{+0.14}_{-0.15}$
121 prop_binCRWS_muon2017_bin2
122 prop_binCRTT_electron2018_bin3
123 prop_binCRTT_electron2017_bin1
124 prop_binCRTT_electron2018_bin4
125 prop_binCRWS_muon2018_bin4
126 QCDScale
127 prop_binCRWS_electron2017_bin3
128 prop_binCRTT_electron2016M_bin1
129 prop_binCRWS_muon2016M_bin6
130 jer_2016M
131 prop_binCRTT_muon2017_bin1
132 jer_2018
133 prop_binSR_electron2016M_bin5
134 prop_binCRTT_muon2016M_bin8
135 puID
136 prop_binSR_electron2018_bin8
137 prop_binCRTT_muon2017_bin5
138 prop_binCRF_electron2017_bin1
139 prop_binSR_electron2017_bin3
140 prop_binCRWS_electron2017_bin2
141 prop_binCRWS_electron2016M_bin6_DYJetsToLL_FxFx
142 prop_binCRWS_electron2016M_bin0
143 prop_binCRF_electron2016M_bin3
144 prop_binCRTT_electron2018_bin2
145 btag
146 prop_binSR_muon2016M_bin1
147 prop_binSR_muon2017_bin3
148 prop_binSR_muon2018_bin8
149 prop_binCRWS_electron2016M_bin2
150 prop_binCRTT_muon2017_bin3
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$

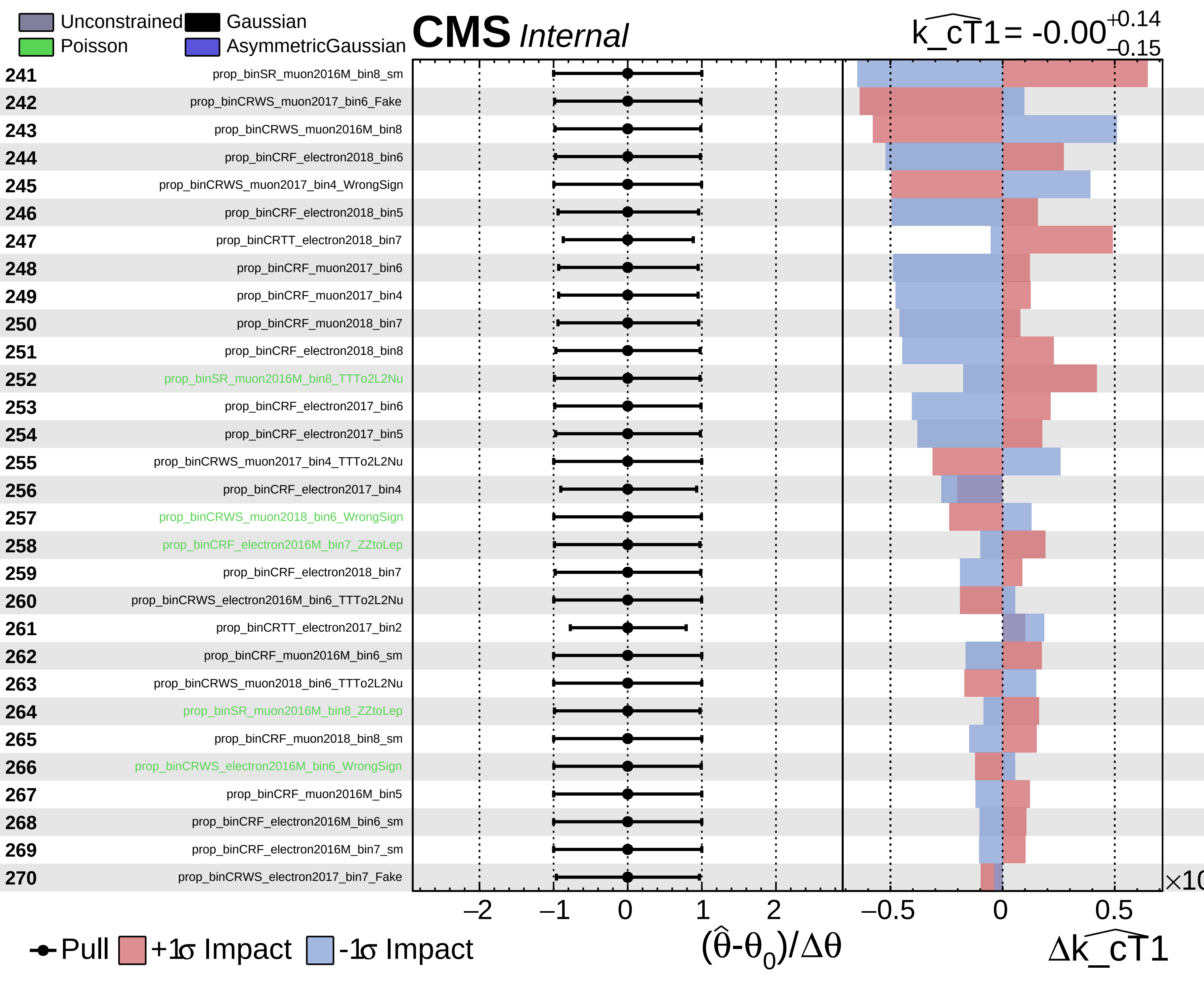


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = -0.00 +0.14 −0.15
151 prop_binCRWS_muon2017_bin4_Fake
152 prop_binCRWS_muon2016M_bin5
153 prop_binCRTT_electron2018_bin8
154 prop_binCRWS_electron2016M_bin8_WrongSign
155 prop_binSR_electron2016M_bin4
156 prop_binCRWS_electron2017_bin6_DYJetsToLL_FxFx
157 prop_binCRTT_electron2016M_bin4
158 prop_binCRTT_electron2017_bin6
159 prop_binCRTT_muon2017_bin7
160 prop_binCRF_muon2018_bin5
161 prop_binCRTT_electron2016M_bin0
162 prop_binCRWS_electron2018_bin5
163 prop_binSR_electron2016M_bin6
164 prop_binCRTT_muon2018_bin5
165 prop_binCRTT_muon2018_bin7
166 prop_binCRWS_electron2018_bin3
167 lumi_2016M
168 prop_binCRF_electron2016M_bin1
169 prop_binCRF_muon2018_bin2
170 tau_vsjet_2017
171 prop_binCRF_muon2016M_bin7
172 prop_binCRWS_electron2017_bin4
173 prop_binCRWS_electron2016M_bin7
174 prop_binCRF_muon2017_bin5
175 prop_binCRF_muon2018_bin6
176 prop_binSR_electron2017_bin8
177 prop_binSR_electron2018_bin2
178 prop_binSR_electron2016M_bin2
179 prop_binCRWS_electron2016M_bin6_Fake
180 prop_binSR_muon2018_bin2
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT1

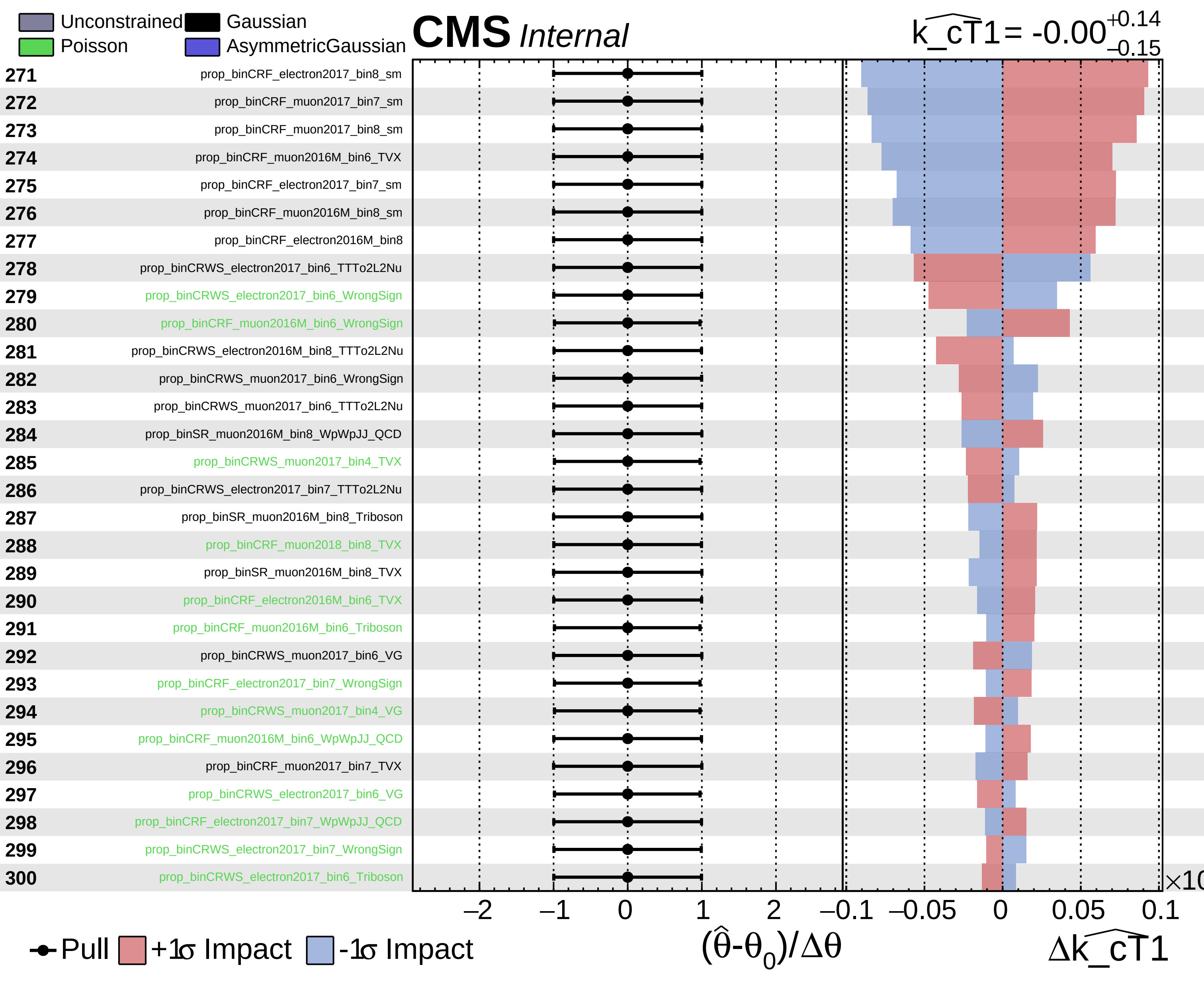

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = -0.00 +0.14 −0.15
181 prop_binCRF_muon2018_bin3
182 prop_binCRTT_muon2018_bin4
183 prop_binCRWS_electron2018_bin7
184 prop_binCRWS_muon2016M_bin7
185 prop_binSR_electron2016M_bin3
186 prop_binCRTT_electron2017_bin3
187 prop_binCRWS_muon2018_bin7
188 lumi_2018
189 prop_binCRWS_muon2017_bin5
190 prop_binCRWS_electron2017_bin7_DYJetsToLL_FxFx
191 prop_binCRF_electron2018_bin4
192 prop_binSR_electron2017_bin2
193 prop_binCRWS_muon2018_bin6_Fake
194 prop_binCRF_muon2016M_bin2
195 prop_binCRWS_electron2016M_bin4
196 prop_binCRWS_muon2018_bin6_DYJetsToLL_FxFx
197 prop_binCRTT_electron2016M_bin5
198 prop_binCRWS_muon2017_bin7
199 prop_binSR_electron2016M_bin1
200 prop_binCRF_muon2017_bin3
201 prop_binCRF_electron2017_bin3
202 prop_binCRF_muon2017_bin2
203 PF
204 prop_binCRF_electron2016M_bin4
205 lumi_2017
206 prop_binSR_muon2016M_bin2
207 prop_binCRTT_muon2017_bin6
208 prop_binCRF_electron2018_bin2
209 prop_binSR_muon2017_bin8
210 prop_binCRTT_electron2016M_bin2
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1



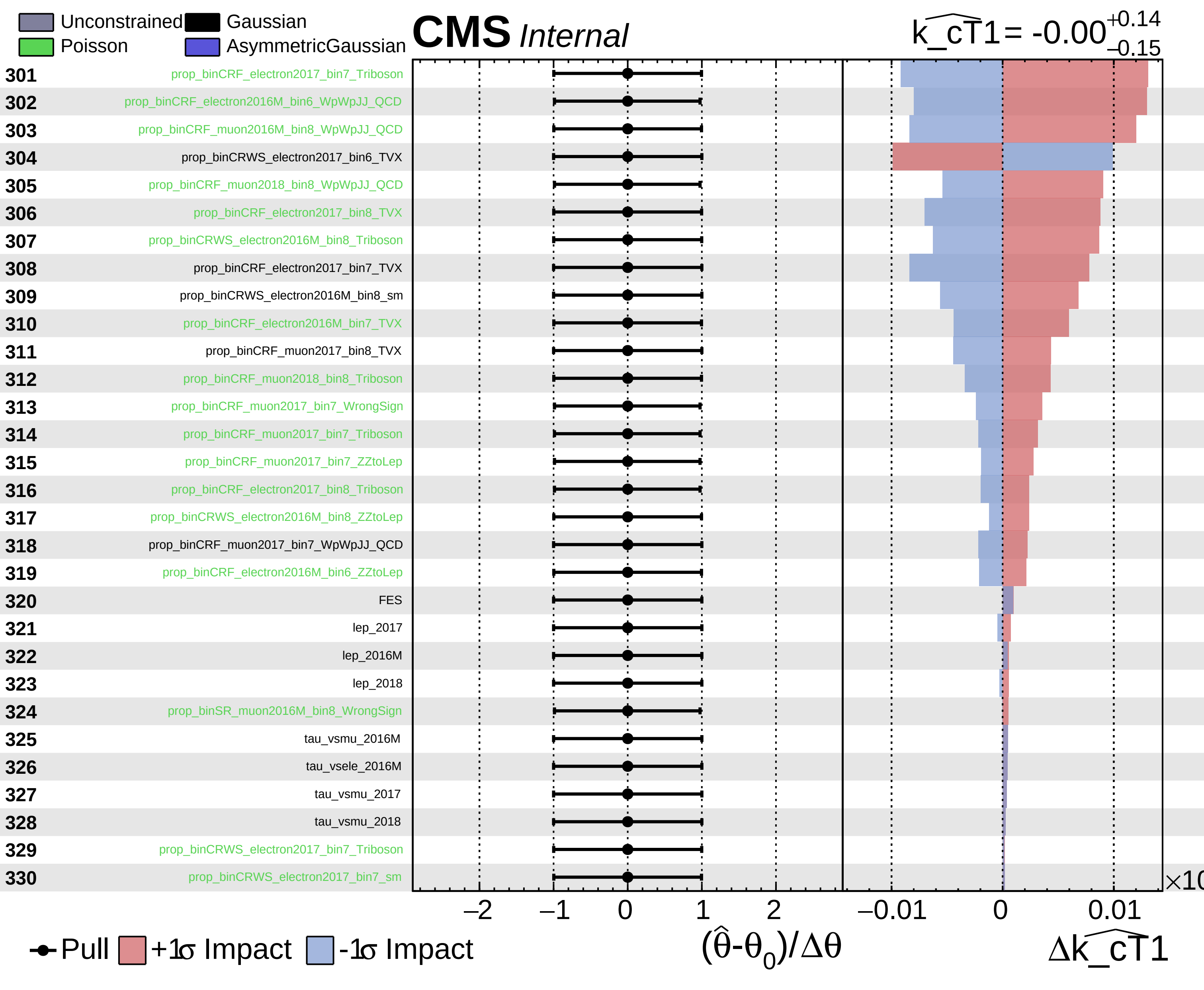
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT1} = -0.00^{+0.14}_{-0.15}$
211 prop_binCRTT_electron2017_bin0
212 prop_binCRF_electron2018_bin3
213 prop_binCRTT_muon2016M_bin6
214 prop_binSR_electron2018_bin1
215 prop_binSR_electron2017_bin1
216 prop_binCRF_electron2016M_bin2
217 tau_vsjet_2018
218 prop_binCRWS_muon2018_bin8
219 prop_binCRF_muon2018_bin4
220 prop_binCRWS_electron2018_bin6
221 prop_binCRWS_electron2016M_bin5
222 prop_binCRTT_electron2017_bin5
223 prop_binCRTT_electron2017_bin4
224 prop_binCRTT_electron2016M_bin7
225 prop_binCRF_electron2016M_bin5
226 prop_binCRTT_electron2017_bin7
227 prop_binCRF_muon2016M_bin3
228 prop_binCRTT_electron2018_bin5
229 prop_binCRF_electron2017_bin2
230 prop_binCRTT_muon2016M_bin4
231 prop_binCRTT_electron2018_bin1
232 prop_binCRF_muon2016M_bin4
233 prop_binCRWS_muon2017_bin8
234 prop_binCRTT_electron2016M_bin3
235 prop_binCRWS_electron2018_bin8
236 prop_binCRF_muon2016M_bin6_ZZtoLep
237 prop_binCRWS_electron2017_bin6_Fake
238 prop_binSR_electron2016M_bin8
239 prop_binCRWS_electron2017_bin8
240 prop_binCRTT_electron2016M_bin6
−2 −1 0 1 2
−0.002 0 0.002
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$



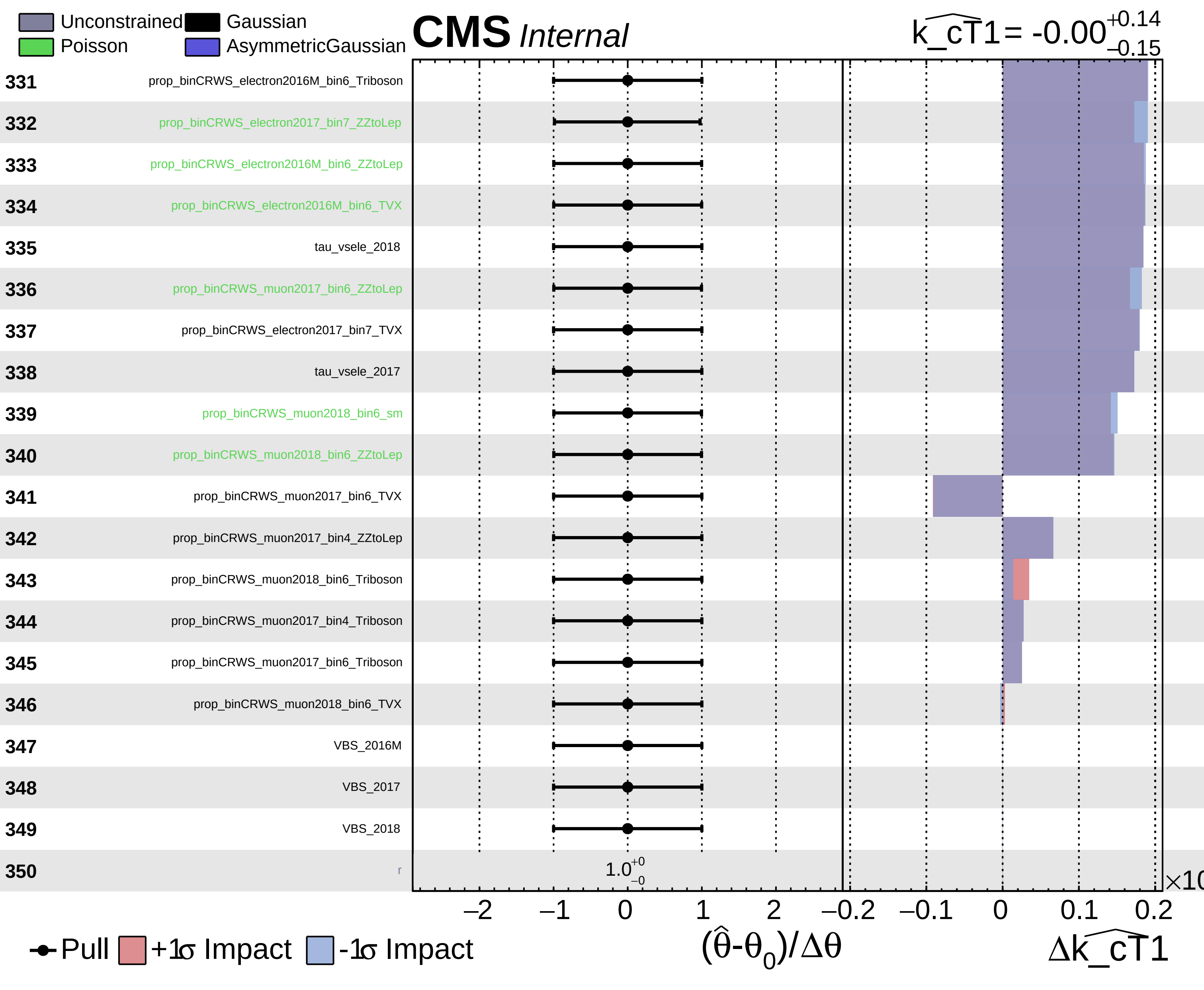
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT1} = -0.00^{+0.14}_{-0.15}$
241 prop_binSR_muon2016M_bin8_sm
242 prop_binCRWS_muon2017_bin6_Fake
243 prop_binCRWS_muon2016M_bin8
244 prop_binCRF_electron2018_bin6
245 prop_binCRWS_muon2017_bin4_WrongSign
246 prop_binCRF_electron2018_bin5
247 prop_binCRTT_electron2018_bin7
248 prop_binCRF_muon2017_bin6
249 prop_binCRF_muon2017_bin4
250 prop_binCRF_muon2018_bin7
251 prop_binCRF_electron2018_bin8
252 prop_binSR_muon2016M_bin8_TTTo2L2Nu
253 prop_binCRF_electron2017_bin6
254 prop_binCRF_electron2017_bin5
255 prop_binCRWS_muon2017_bin4_TTTo2L2Nu
256 prop_binCRF_electron2017_bin4
257 prop_binCRWS_muon2018_bin6_WrongSign
258 prop_binCRF_electron2016M_bin7_ZZtoLep
259 prop_binCRF_electron2018_bin7
260 prop_binCRWS_electron2016M_bin6_TTTo2L2Nu
261 prop_binCRTT_electron2017_bin2
262 prop_binCRF_muon2016M_bin6_sm
263 prop_binCRWS_muon2018_bin6_TTTo2L2Nu
264 prop_binSR_muon2016M_bin8_ZZtoLep
265 prop_binCRF_muon2018_bin8_sm
266 prop_binCRWS_electron2016M_bin6_WrongSign
267 prop_binCRF_muon2016M_bin5
268 prop_binCRF_electron2016M_bin6_sm
269 prop_binCRF_electron2016M_bin7_sm
270 prop_binCRWS_electron2017_bin7_Fake
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT1} = -0.00^{+0.14}_{-0.15}$
271 prop_binCRF_electron2017_bin8_sm
272 prop_binCRF_muon2017_bin7_sm
273 prop_binCRF_muon2017_bin8_sm
274 prop_binCRF_muon2016M_bin6_TVX
275 prop_binCRF_electron2017_bin7_sm
276 prop_binCRF_muon2016M_bin8_sm
277 prop_binCRF_electron2016M_bin8
278 prop_binCRWS_electron2017_bin6_TTTo2L2Nu
279 prop_binCRWS_electron2017_bin6_WrongSign
280 prop_binCRF_muon2016M_bin6_WrongSign
281 prop_binCRWS_electron2016M_bin8_TTTo2L2Nu
282 prop_binCRWS_muon2017_bin6_WrongSign
283 prop_binCRWS_muon2017_bin6_TTTo2L2Nu
284 prop_binSR_muon2016M_bin8_WpWpJJ_QCD
285 prop_binCRWS_muon2017_bin4_TVX
286 prop_binCRWS_electron2017_bin7_TTTo2L2Nu
287 prop_binSR_muon2016M_bin8_Triboson
288 prop_binCRF_muon2018_bin8_TVX
289 prop_binSR_muon2016M_bin8_TVX
290 prop_binCRF_electron2016M_bin6_TVX
291 prop_binCRF_muon2016M_bin6_Triboson
292 prop_binCRWS_muon2017_bin6_VG
293 prop_binCRF_electron2017_bin7_WrongSign
294 prop_binCRWS_muon2017_bin4_VG
295 prop_binCRF_muon2016M_bin6_WpWpJJ_QCD
296 prop_binCRF_muon2017_bin7_TVX
297 prop_binCRWS_electron2017_bin6_VG
298 prop_binCRF_electron2017_bin7_WpWpJJ_QCD
299 prop_binCRWS_electron2017_bin7_WrongSign
300 prop_binCRWS_electron2017_bin6_Triboson
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT1}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = -0.00 +0.14 −0.15
301 prop_binCRF_electron2017_bin7_Triboson
302 prop_binCRF_electron2016M_bin6_WpWpJJ_QCD
303 prop_binCRF_muon2016M_bin8_WpWpJJ_QCD
304 prop_binCRWS_electron2017_bin6_TVX
305 prop_binCRF_muon2018_bin8_WpWpJJ_QCD
306 prop_binCRF_electron2017_bin8_TVX
307 prop_binCRWS_electron2016M_bin8_Triboson
308 prop_binCRF_electron2017_bin7_TVX
309 prop_binCRWS_electron2016M_bin8_sm
310 prop_binCRF_electron2016M_bin7_TVX
311 prop_binCRF_muon2017_bin8_TVX
312 prop_binCRF_muon2018_bin8_Triboson
313 prop_binCRF_muon2017_bin7_WrongSign
314 prop_binCRF_muon2017_bin7_Triboson
315 prop_binCRF_muon2017_bin7_ZZtoLep
316 prop_binCRF_electron2017_bin8_Triboson
317 prop_binCRWS_electron2016M_bin8_ZZtoLep
318 prop_binCRF_muon2017_bin7_WpWpJJ_QCD
319 prop_binCRF_electron2016M_bin6_ZZtoLep
320 FES
321 lep_2017
322 lep_2016M
323 lep_2018
324 prop_binSR_muon2016M_bin8_WrongSign
325 tau_vsmu_2016M
326 tau_vsele_2016M
327 tau_vsmu_2017
328 tau_vsmu_2018
329 prop_binCRWS_electron2017_bin7_Triboson
330 prop_binCRWS_electron2017_bin7_sm
−2 −1 0 1 2
−0.01 0 0.01
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT1 = -0.00 +0.14 −0.15
331 prop_binCRWS_electron2016M_bin6_Triboson
332 prop_binCRWS_electron2017_bin7_ZZtoLep
333 prop_binCRWS_electron2016M_bin6_ZZtoLep
334 prop_binCRWS_electron2016M_bin6_TVX
335 tau_vsele_2018
336 prop_binCRWS_muon2017_bin6_ZZtoLep
337 prop_binCRWS_electron2017_bin7_TVX
338 tau_vsele_2017
339 prop_binCRWS_muon2018_bin6_sm
340 prop_binCRWS_muon2018_bin6_ZZtoLep
341 prop_binCRWS_muon2017_bin6_TVX
342 prop_binCRWS_muon2017_bin4_ZZtoLep
343 prop_binCRWS_muon2018_bin6_Triboson
344 prop_binCRWS_muon2017_bin4_Triboson
345 prop_binCRWS_muon2017_bin6_Triboson
346 prop_binCRWS_muon2018_bin6_TVX
347 VBS_2016M
348 VBS_2017
349 VBS_2018
350 r
1.0 +0 −0
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cT1